

JACOBABAD, Pakistan

WMO: 417150

Lat: 28.284N

Lon: 68.450E

Elev: 185

StdP: 14.60

Time Zone: 5.00 (E05)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.7	47.0	25.4	19.5	67.1	30.0	24.3	66.7	10.6	61.3	8.6	64.5	0.5	50	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	21.9	115.4	78.2	112.4	77.6	109.7	78.5	88.2	102.2	86.4	99.8	85.3	98.8	3.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
85.3	187.3	97.1	83.3	175.3	95.6	82.1	168.5	94.8	54.0	102.1	51.6	100.0	50.3	98.9	100.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.2	8.9	7.0	40.2	120.4	3.8	1.5	37.5	121.5	35.3	122.4	33.2	123.2	30.4	124.4	
			DB												
			WB	36.9	91.3	3.9	3.5	34.1	93.9	31.8	95.9	29.7	97.9	26.8	100.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.7	59.2	65.4	76.2	89.0	97.6	98.6	95.4	91.8	89.0	82.7	72.2	62.1
	DBStd	14.23	3.80	5.12	6.42	5.65	4.40	3.38	3.17	2.49	2.60	3.77	4.42	4.17
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD65	347	182	52	3	0	0	0	0	0	0	0	1	109
	CDD50	11559	286	431	811	1169	1475	1459	1407	1295	1169	1015	667	375
	CDD65	6432	3	63	350	719	1010	1009	942	830	719	550	218	19
	CDH74	97837	37	541	3732	10822	17248	17592	15694	13121	10599	6459	1804	188
	CDH80	65122	1	116	1762	7047	12814	13278	11234	8666	6328	3283	577	16
Wind	WSAvg	1.9	1.0	1.7	1.8	2.2	2.8	3.5	2.8	2.2	1.7	1.3	0.8	0.8
Precipitation	PrecAvg	21.2	3.6	2.6	2.6	1.1	0.5	0.4	0.1	0.2	0.5	1.0	2.5	2.8
	PrecMax	50.3	17.0	9.9	11.7	5.8	3.2	2.2	1.0	1.5	7.3	9.9	8.9	10.4
	PrecMin	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	12.0	3.9	2.9	3.0	1.4	0.8	0.6	0.2	0.3	1.1	1.6	2.9	2.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.9	86.8	102.4	113.0	119.9	119.9	111.2	103.4	102.0	98.8	92.1	82.6
		MCWB	59.8	65.0	71.7	73.4	76.3	79.2	84.9	83.8	82.3	77.6	69.1	64.8
	2%	DB	74.4	83.6	97.3	109.7	116.4	115.4	107.9	101.4	99.4	96.5	88.5	78.8
		MCWB	58.9	65.2	70.0	73.0	77.0	79.9	84.4	84.1	81.8	77.6	69.6	63.1
	5%	DB	71.8	80.4	92.8	106.4	113.3	112.1	105.7	100.0	97.7	94.5	86.1	76.0
		MCWB	58.0	63.6	69.9	73.0	76.7	79.7	84.3	84.0	81.6	76.4	68.0	61.3
	10%	DB	69.5	77.2	89.2	102.8	110.5	109.5	103.5	98.5	96.1	92.7	83.7	73.3
		MCWB	57.3	61.9	68.9	72.5	76.4	81.0	84.1	83.9	81.6	75.3	67.3	59.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.1	69.3	79.2	84.5	88.7	96.7	93.3	87.8	87.0	82.7	75.5	67.6
		MCDB	73.2	81.9	93.1	103.5	109.7	109.3	101.2	98.0	97.3	93.3	86.2	78.7
	2%	WB	61.0	67.2	74.3	79.6	85.1	88.8	87.9	86.1	85.3	81.2	72.9	65.2
		MCDB	71.2	79.7	89.5	98.8	105.3	105.3	100.6	97.3	95.6	91.8	84.0	76.0
	5%	WB	59.7	65.4	71.9	77.0	82.5	85.4	86.3	85.2	84.2	80.0	71.1	63.1
		MCDB	69.5	77.9	87.5	96.8	103.0	103.0	99.8	96.6	94.1	90.5	81.9	73.0
	10%	WB	58.3	63.5	70.2	75.2	80.4	84.0	85.4	84.5	83.1	78.8	69.3	61.3
		MCDB	67.7	74.9	85.8	96.6	101.8	102.9	99.3	95.9	93.0	89.0	80.1	71.0
Mean Daily Temperature Range	5% DB	MDBR	20.8	21.4	22.4	24.9	25.4	21.9	16.4	13.6	14.4	19.6	22.1	21.9
		MCDBR	23.9	25.1	26.4	29.3	29.3	26.8	19.5	15.3	16.3	20.8	24.1	24.5
		MCWBR	12.0	11.9	9.9	10.0	9.6	8.4	6.1	5.4	5.8	7.5	9.5	12.2
	5% WB	MCDBR	21.4	22.2	23.8	25.7	26.5	21.7	17.0	14.2	14.4	17.2	20.4	21.8
		MCWBR	12.2	11.6	11.6	13.1	12.2	8.8	6.7	6.0	6.7	7.8	10.0	12.2
Clear-Sky Solar Irradiance	taub	0.487	0.502	0.535	0.595	0.627	0.714	0.814	0.733	0.594	0.542	0.568	0.512	
	taud	1.855	1.862	1.800	1.710	1.668	1.559	1.453	1.579	1.788	1.790	1.705	1.807	
	Ebn at Noon	229	240	242	232	225	205	185	199	224	225	202	213	
	Edh at Noon	56	60	68	76	79	88	98	86	67	63	64	56	
All-Sky Solar Radiation	RadAvg	1199	1466	1841	2141	2336	2268	2103	2015	1958	1710	1321	1138	
	RadStd	70	109	89	58	47	64	100	116	82	60	61	53	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page